A Native Spin-Polarized Fusion Neutron Source for OpenMC: Stellarator Wall-Loading, the Rate–Steering Trade-off, and Where Monte-Carlo Transport Is Necessary

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Abstract

Spin polarization of deuterium–tritium fuel both enhances the fusion rate and makes the 14.1 MeV neutron emission anisotropic about the local magnetic field, so the first-wall neutron load of a polarized reactor depends on the plasma shape, the field geometry, and the polarization mix together. We build the first *native, on-the-fly* spin-polarized fusion neutron *source* in a production Monte-Carlo transport code (OpenMC): a thread-safe sampler that draws the polarized birth direction exactly, for an arbitrary polarization mix and an arbitrary *local* field direction evaluated per birth point. Because the local field is required at every position, the source is realized as a `StellaratorSource` that draws births and the unit field $\hat{\boldsymbol{B}}$ from a verified three-dimensional VMEC equilibrium—the first equilibrium-rigorous, polarized stellarator source, and the first stellarator source of any kind in OpenMC. We give the emission kernel and its separation into a birth-direction distribution and a scalar rate factor, an exact stratified sampler with a closed-form inverse-CDF cross-checked against rejection, and a degeneracy-free frame rotation; we verify the sampler in isolation to 0.3% per bin, and verify the transported loads against the analytic free-streaming NWL where the two must agree ($\leq$ 1.8% on a real published QA equilibrium, with a 4–6% inboard residual that we identify by construction as the analytic point-patch limit, not a defect of the source). We then use the source to do what the free-streaming and analytic pictures cannot. First, *scattering dilutes the geometric steering by a factor of* $\sim$3: the $\pm$40% inboard contrast of the free-streaming limit realizes as only $\pm$13–15% through a real blanket, because backscattered neutrons return as a direction-blind diffuse flux. Second, restoring the reactivity separates what the three polarization modes buy—the +50%-rate mode compounds the center-stack load to 1.70×, while the equal-rate parallel mode relieves it to 0.85× at no cost to power or to the tritium breeding ratio. Third, on conformal first walls of real quasi-symmetric stellarators, peaking-optimal polarization lowers the peaking factor only modestly (e.g. 1.78 → 1.49): the load peak is a narrow near-field spike that angular redistribution cannot move, so polarization is a strong *local* actuator but a weak *global* flattener—a conclusion an independent differentiable analytic method reaches as well. The source is available for both tokamak- and stellarator-class equilibria and provides a verified foundation for polarized-fuel reactor neutronics.

1 Introduction

Spin-polarized fusion (SPF) aligns the reactant nuclear spins to raise the D–T reaction cross section by up to 50% and to redirect the angular distribution of the fusion products [4, 3, 1]. The neutron carries 80% of the D–T energy, so the redistribution of the 14.1 MeV neutrons directly reshapes the first-wall neutron load—the quantity that sets the blanket, shield, and material lifetimes of a reactor, and the second of the two central problems of fusion neutronics alongside tritium self-sufficiency [6]—and the optimal polarization choice is a system decision that must be made against that load [1, 2]. Scoping studies quantify the load through the neutron wall load (NWL), the first-flight neutron current through a wall patch weighted by the neutron energy; it omits scattering but is inexpensive and is therefore the natural object for design iteration [2, 1]. A reactor calculation, however, needs the transported load: the volumetric heating and breeding that follow once the anisotropic birth distribution is carried through a real, scattering blanket.

The polarized neutron emission is not isotropic. For a chosen mixture of deuteron and triton spin projections the differential cross section is a fixed combination of two angular shapes referred to the local field direction $\hat{\boldsymbol{B}}$: a $\sin^2\theta$ shape that peaks perpendicular to $\hat{\boldsymbol{B}}$ and a $\frac{1}{4}+\frac{3}{4}\cos^2\theta$ shape that peaks parallel to it, where θ is the angle between the emitted neutron and $\hat{\boldsymbol{B}}$ [1]. Because $\hat{\boldsymbol{B}}$ follows the magnetic geometry, the wall load of a polarized plasma is a convolution of the emission kernel, the field, and the source distribution over the plasma volume against the wall geometry—and evaluating the emission at the *local* field direction is precisely what a shape parametrization cannot supply and a solved equilibrium can.

Prior work, and the gap. Two complementary routes evaluate the polarized NWL convolution. The first is analytic. For an axisymmetric device the toroidal source integral over a planar ring reduces to incomplete elliptic integrals, giving a fast, automatically differentiable NWL suited to shape and polarization optimization [1]; a companion analytic development (§10) restores this closed form perturbatively for weakly shaped QA boundaries, valid while the geometric non-axisymmetry is small, and is free-streaming by construction. The second route is Monte-Carlo, and here the closest prior work must be placed carefully. [5] performed the first polarized-source neutronics in OpenMC, but by *precomputing* a static point-source cloud for one fixed spherical- tokamak scheme; the emission directions are baked in ahead of the run rather than sampled on the fly in the local field frame, which is what a spatially varying stellarator field requires. Parametric stellarator neutronics in production codes has meanwhile focused on the *geometry*: [6] scan blanket thickness for a HELIAS in Serpent2 with a neutron source "integrated ... as a user-defined source routine based on the MCNP code"—a ported, isotropic routine, not one derived from the equilibrium's own volume element—and parametric tokamak plasma sources exist in OpenMC as well [7], but are unpolarized and axisymmetric. What does not yet exist is a *native, general, on-the-fly polarized* source that both samples the correct anisotropic emission in the local field frame and is driven by a real three-dimensional equilibrium so that it applies to stellarators. We build it.

Contributions and organization. We contribute (i) the first native, on-the-fly polarized fusion source in a production MC code, and its verification, including an exact self-consistency identity in the polarization parameter; (ii) `StellaratorSource`, equilibrium-rigorous polarized emission from a gated VMEC equilibrium, the first such source in OpenMC; and (iii) the use of that source to establish results the free-streaming and analytic pictures structurally cannot—the $\sim 3\times$ scattering dilution of steering, the rate-versus-steering trade-off among the polarization modes, and the finding that polarization is a strong local but weak global actuator of the wall load, the last cross-validated by an independent analytic method. The paper is organized as a ladder of models, each used at the fidelity that isolates one effect. Section 2 states the emission kernel and its birth-direction/rate separation. Section 3 gives the exact sampler, the equilibrium birth-position and field models, and the OpenMC realization. Section 4 verifies the sampler in isolation. Section 5 cross-validates the transported free-streaming loads against the analytic method on a real published QA equilibrium, and Section 6 identifies the one residual as the analytic point-patch limit. Section 7 turns on scattering and reports the dilution of the steering; Section 8 restores the reactivity and the rate-versus-steering trade-off; Section 9 evaluates conformal-wall loads for real stellarators and the local-versus-global result. Section 10 describes the analytic companion and the division of labour, Section 11 bounds the claims, and Section 12 summarizes.

2 The spin-polarized emission source

We follow the parameterization of Ref. [1]. Let the deuteron spin projection fractions be d_+, d_0, d_- and the triton fractions t_+, t_-. The three collision-mode fractions are

$$a = d_+t_+ + d_-t_-, \qquad b = d_0, \qquad c = d_+t_- + d_-t_+, \qquad a+b+c = 1, \tag{1}$$

with the unpolarized fuel at $a = b = c = \frac{1}{3}$.[1] The differential cross section, referred to the local field direction $\hat{\boldsymbol{B}}$, is

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{\sigma_0}{2\pi}\left[\tfrac{3}{4}\,a\,\sin^2\theta + \left(\tfrac{2}{3}b + \tfrac{1}{3}c\right)\left(\tfrac{1}{4} + \tfrac{3}{4}\cos^2\theta\right)\right], \qquad \cos\theta = \hat{\boldsymbol{B}}\cdot\hat{\boldsymbol{\Delta}}, \tag{2}$$

where $\hat{\boldsymbol{\Delta}}$ is the emitted neutron direction and σ_0 is the cross section in the $S = \frac{3}{2}$ combined-spin state. Integrating Eq. (2) over the sphere gives the total rate $\sigma_{\text{tot}} = \sigma_0\,\eta$ with $\eta = a + \frac{2}{3}b + \frac{1}{3}c$; the unpolarized value is $\frac{2}{3}\sigma_0$, pure a (mode A) is σ_0 (a 50% enhancement), and pure c (mode C) is $\frac{1}{3}\sigma_0$. Modes B and C share the same angular shape $\frac{1}{4} + \frac{3}{4}\cos^2\theta$ and differ only in rate: the emission of a polarized plasma is controlled by two angular functions, not three. (Mode C is therefore *not* isotropic—a common misassignment—but shares its directionality with mode B while emitting at half the rate.)

The essential structure for sampling is that the *birth direction* and the *total rate* separate. Collecting the two angular shapes,

$$I(\theta) = w_\perp \sin^2\theta + w_\parallel\left(\tfrac{1}{4} + \tfrac{3}{4}\cos^2\theta\right), \qquad w_\perp = \tfrac{3}{4}a, \quad w_\parallel = \tfrac{2}{3}b + \tfrac{1}{3}c, \tag{3}$$

the normalized birth-direction density is $p(\theta,\phi) = I(\theta)/\int I\,\mathrm{d}\Omega$ with ϕ uniform, and it depends only on $(w_\perp, w_\parallel)$. The overall rate factor η multiplies the spatial source strength and does not enter the direction sampling. Keeping these two concerns separate is the central design choice: the sampler draws a correctly shaped direction for any (a,b,c), and η is applied, if absolute loads are wanted, as a scalar weight on the birth-position distribution. It is convenient to summarize the direction by a single parameter, writing the normalized intensity as $w(\theta) \propto 1 + a_2\,P_2(\cos\theta)$ with $a_2 \in [-1,+1]$, where $a_2 = -1$ is the pure perpendicular shape, $a_2 = +1$ the pure parallel shape, and $a_2 = 0$ isotropic; because P_2 enters linearly, the wall load is exactly linear in a_2, a fact we exploit for verification in §9. For the free-streaming cross-checks the direction is all that matters, and the birth energy is taken monoenergetic at 14.1 MeV; a Ballabio-broadened Gaussian at a specified ion temperature is available for physical runs and is irrelevant to the angular verification.

[1] Our (a,b,c) are Schwartz's collision-mode fractions [1]; they should not be confused with the identically-named vector/tensor polarization parameters used by Bae [5], whose scheme has no separate "mode c."

3 Monte-Carlo realization

Exact stratified direction sampling. Because Eq. (3) is a nonnegative combination of two shapes, we sample the birth direction by stratification: with probability $W_\perp/(W_\perp+W_\parallel)$ draw from the $\sin^2\theta$ shape, otherwise from the $\frac{1}{4}+\frac{3}{4}\cos^2\theta$ shape, where $W_\perp = w_\perp \int \sin^2\theta\,\mathrm{d}\Omega = \frac{8\pi}{3}w_\perp$ and $W_\parallel = w_\parallel \int (\frac{1}{4}+\frac{3}{4}\cos^2\theta)\,\mathrm{d}\Omega = 2\pi w_\parallel$ are the solid-angle-integrated weights. This is an exact draw from the mixture and makes per-mode verification trivial. Each shape is sampled by a closed-form inverse-CDF in $x=\cos\theta$: the $\sin^2\theta$ shape has marginal $\propto \sin^3\theta$ and is inverted by the trigonometric solution of the depressed cubic $x^3-3x+2(1-2\xi)=0$; the $\frac{1}{4}+\frac{3}{4}\cos^2\theta$ shape has marginal $\propto \frac{1}{4}+\frac{3}{4}x^2$ and is a cubic in x as well. Each inverse-CDF is cross-checked against an independent rejection sampler (envelope $\propto \sin\theta$, acceptance $\frac{2}{3}$) to a two-sample Kolmogorov–Smirnov tolerance, so the root selection in the cubic is verified rather than assumed. The azimuth ϕ is uniform on $[0,2\pi)$. The sampler holds no mutable state: every derived weight is fixed at construction and the sampling methods are constant, so the concurrent, per-thread calls OpenMC makes are race-free, and randomness comes only from OpenMC's own counter-based stream.

Local-to-global rotation. The sampled direction is expressed in a frame with $\hat{\boldsymbol{z}}_{\mathrm{loc}} \parallel \hat{\boldsymbol{B}}$ and rotated to global Cartesian by a Gram–Schmidt construction that avoids the $\hat{\boldsymbol{B}} \parallel \hat{\boldsymbol{z}}$ degeneracy: the reference axis is whichever of $\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}, \hat{\boldsymbol{z}}$ is least aligned with $\hat{\boldsymbol{B}}$, and $\hat{\boldsymbol{x}}_{\mathrm{loc}} = \mathrm{normalize}(\hat{\boldsymbol{e}}_{\mathrm{ref}} - (\hat{\boldsymbol{e}}_{\mathrm{ref}}\cdot\hat{\boldsymbol{B}})\hat{\boldsymbol{B}})$, $\hat{\boldsymbol{y}}_{\mathrm{loc}} = \hat{\boldsymbol{B}} \times \hat{\boldsymbol{x}}_{\mathrm{loc}}$. The resulting global direction is unit to machine precision and, tested with a random non-axis $\hat{\boldsymbol{B}}$, leaves an isotropic local distribution isotropic—the check that the rotation carries no bias.

Birth positions and the equilibrium field. The birth-position model is pluggable. For the free-streaming cross-checks the plasma is represented by constant-(ρ,ϑ) filaments on a set of scaled flux surfaces, sampled uniformly in the toroidal angle and weighted by the arc-length Jacobian and a model emissivity $W(\rho)\propto\rho^2$; the same filaments are the source loops of the analytic method, so the two calculations share the source exactly and the comparison isolates the transport treatment. For the equilibrium-driven runs of §9, births are drawn from a three-dimensional VMEC [8] equilibrium with the physically correct weight, reactivity times the metric Jacobian $\sqrt{g}$, and the unit field is reconstructed from the equilibrium as $\boldsymbol{B} = B^u\partial_\vartheta\boldsymbol{x} + B^v\partial_\zeta\boldsymbol{x}$ with $\hat{\boldsymbol{B}} = \boldsymbol{B}/|\boldsymbol{B}|$. The producer hard-gates its invariants—an independent divergence-theorem volume must match the VMEC volume to better than 1.5×10^{-2}, and $\sqrt{g}$ must be single-signed—so the spatial distribution and the field direction the source consumes are verified, not assumed. This upgrades the parametric tokamak-source pattern [7] on four axes: real equilibrium geometry, correct $\sqrt{g}$ volumetric weighting, genuine three-dimensionality, and the local field direction that makes the polarized emission well posed in the first place.

OpenMC source and transport. The sampled births are handed to OpenMC [11] either through a compiled custom-source shared library or as a pre-sampled source bank written to an HDF5 file and read back as a fixed source. For the free-streaming cross-checks the vessel interior is void and the walls are vacuum boundaries, so each neutron streams in a straight line from its birth point until it crosses a surface and is tallied; the wall load is a surface-current tally binned poloidally. This ray-traced transport makes no point-patch or $1/D^2$ approximation—a neutron reaches a wall patch if and only if its sampled trajectory does—which is the property that the analytic comparison below tests. The same source, run with a material first wall and blanket rather than a void, gives the transported load (§7–9); on the conformal geometries the first wall is a CAD

Table 1: Sampler self-test: sampled directionality A/iso and B/iso per $\cos\theta$ bin against the exact normalized kernels, and the mode means. The sampled kernels match to 0.3% and integrate to unit emission (1.0000), so the direction draw is exact independent of any geometry.

$\cos\theta$	A/iso (MC)	$\frac{3}{2}\sin^2\theta$	B/iso (MC)	$2(\frac{1}{4}+\frac{3}{4}\cos^2\theta)$
−0.95	0.146	0.146	1.857	1.854
−0.15	1.467	1.466	0.535	0.534
+0.05	1.491	1.496	0.505	0.504
+0.85	0.414	0.416	1.582	1.584
mean	1.0000	1	1.0000	1

surface imported through DAGMC [12].

4 Verification of the sampler

The sampler is verified before any geometry is introduced. Drawing 2×10^6 directions per mode about a fixed $\hat{\boldsymbol{B}}=\hat{\boldsymbol{z}}$ and histogramming in equal $\cos\theta$ bins, the sampled directionality—the ratio of the mode density to the isotropic density in each bin—reproduces the normalized kernels $\frac{3}{2}\sin^2\theta$ (mode A) and $2(\frac{1}{4}+\frac{3}{4}\cos^2\theta)$ (mode B/C) to better than 0.3% per bin, and the mean directionality of each polarized mode is 1.0000, confirming that the stratified draw is normalized to unit emission exactly (Table 1). A χ^2 test against the analytic $p(\cos\theta)$ for {unpolarized, A, B, C, mixed} and a two-sample KS test of inverse-CDF against rejection close the sampler verification; the isotropic and rotation-isotropy checks are flat as required. This isolates the sampler from every subsequent test: any later discrepancy is geometric, not a defect of the angular draw.

5 Cross-validation against the analytic method

The decisive test is that the Monte-Carlo source reproduces the analytic free-streaming NWL where the analytic method is itself exact. We compare on a shared square-cross-section torus wall, so that all non-axisymmetry lives in the source and the wall is common to both calculations, and we compare the wall directionality A/iso and B/iso—the ratio of the polarized load to the isotropic load on each wall—which cancels the source normalization and isolates the polarized steering. The analytic side is a pure-numpy free-streaming quadrature that evaluates the loop geometry and field directly at the Gauss–Legendre nodes with the true shaped-loop visibility and the inner-cylinder occlusion; it is gated at run time against the companion `anarríma` quadrature and matches it to 10^{-12}, so it is the analytic method's own reference, only faster. We use a genuine published equilibrium: because the standard published QA configurations are all strongly shaped (a survey of DESC examples gives $\varepsilon_{\text{eff}}\approx 1$–2.6 at a tight wall), we mined the QUASR stellarator database [13] for a gentle one without the bulk download—pulling its single tabulated catalogue (3.7×10^5 devices), filtering to quasi-axisymmetry, ranking by a metadata proxy that correlates with the geometric excursion (correlation 0.94), and computing the exact ε_{eff} of a shortlist from the VMEC boundary spectra. The gentlest is a $N_{\text{fp}}=3$ QA of aspect ratio 6.7, which at the validation wall has $\varepsilon_{\text{eff}}\approx0.43$—four times more shaped than an idealized toy and, unlike it, near the edge of the analytic method's convergent band (Fig. 1). The published QAs proper sit well beyond that edge: at their native $\varepsilon_{\text{eff}}\approx2.3$ the analytic series diverges and even the fixed-node exact quadrature fails, disagreeing with the reference by $\sim$34% and returning unphysical negative loads from the near-singular grazing

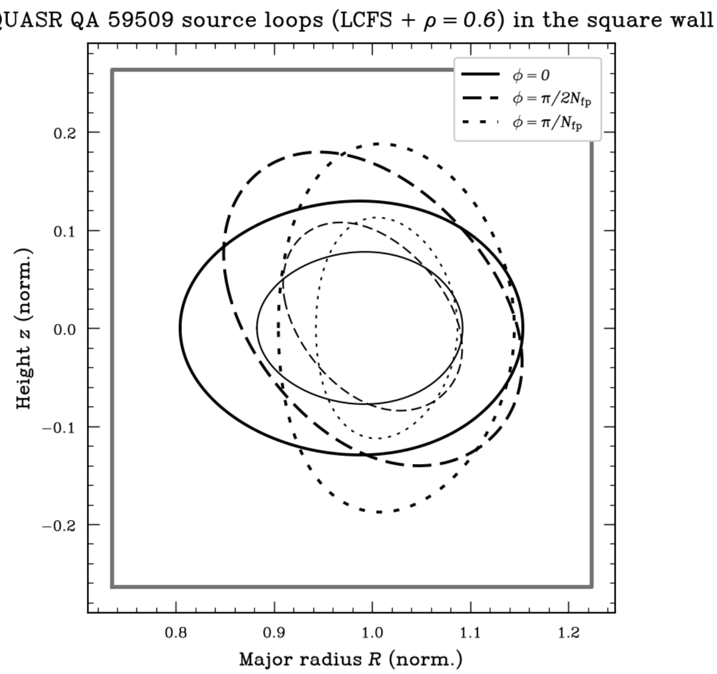

Figure 1: Source loops of the real QUASR QA equilibrium (device 59509, $N_{\mathrm{fp}} = 3$): the last-closed and an interior flux surface at several toroidal cuts, showing the rotating, strongly shaped cross section, inside the square-cross-section wall used for the cross-check.

Table 2: Real QUASR QA ($\varepsilon_{\mathrm{eff}} \approx 0.43$): directionality A/iso, B/iso as a ratio of total wall loads, Monte-Carlo vs analytic. Outboard/floor/ceiling agree to $\leq 1.8\%$; the inboard wall carries the 4–6% residual analyzed in §6.

wall	A MC	A an	B MC	B an
outboard	0.833	0.848	1.165	1.152
floor	1.048	1.057	0.955	0.943
ceiling	1.043	1.057	0.954	0.943
inboard	1.248	1.199	0.755	0.801

integrand—so the strongly shaped configurations are Monte-Carlo-only, and it is the gentle mined equilibrium that permits a cross-check at all.

On this configuration the Monte-Carlo and analytic directionalities agree to $\leq 1.8\%$ on the outboard, floor, and ceiling walls, with the poloidal profiles tracked to correlation 0.7–0.99; the inner (inboard) wall shows a larger, 4–6% residual (Table 2, Fig. 2). The inboard wall is where the analytic companion's own controlling parameter $\varepsilon_{\mathrm{eff}}$ is largest—it is local and peaks where the standoff is smallest—so a departure there, on the most strongly curved wall of the most strongly shaped case, is exactly where one expects the free-streaming analytic to be most stressed. The next section shows that this residual is a limit of the analytic method, not of the Monte-Carlo source.

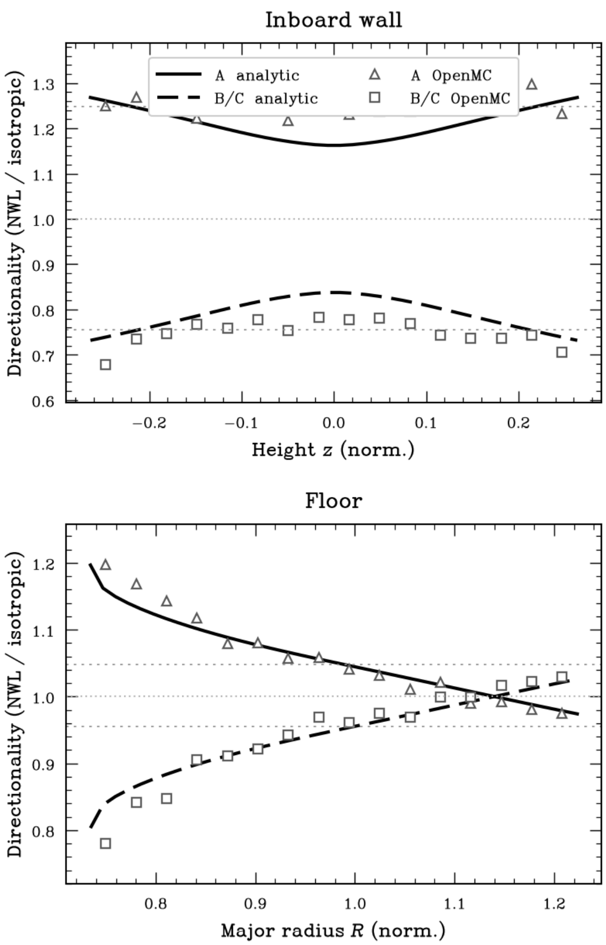

Figure 2: Free-streaming directionality A/iso (mode A) and B/iso (mode B/C) around each wall of the square torus for the real QUASR QA equilibrium ($\varepsilon_{\text{eff}} \approx 0.43$): the analytic point-patch flux integral (curves) and the OpenMC ray-traced source (markers). The outboard, floor, and ceiling walls overlay; the inboard wall tracks the poloidal shape with the 4–6% offset. Per-panel vertical scales because the walls differ by an order of magnitude in poloidal curvature.

6 The point-patch limit of the analytic method

The inboard residual is small but coherent, and its cause is worth establishing because it maps the boundary between the two methods. We first eliminate the Monte-Carlo source as the cause, then reproduce the effect by direct construction. The residual is not the direction sampler, which reproduces the kernels exactly (§4); not the analytic quadrature, which matches `anarrīma` to 10^{-12}; not the visibility model, since using the true shaped-loop sub-arcs with inner-cylinder occlusion rather than a mean-radius arc removes an unrelated outboard error but leaves the inboard value unchanged; not the field model, since the analytic and Monte-Carlo evaluations call the identical $\hat{\boldsymbol{B}}$, which cannot by itself open a gap between them; and not a near-field effect, since moving the wall outward from a gap of $0.4\,a$ to $0.7\,a$ leaves the offset unchanged ($0.036 \to 0.034$). What does change it is the plasma shaping—the gentle case agrees on the inboard wall, the strongly shaped one does not—so the effect is a coupling of the source shaping to the wall geometry that the two methods treat differently.

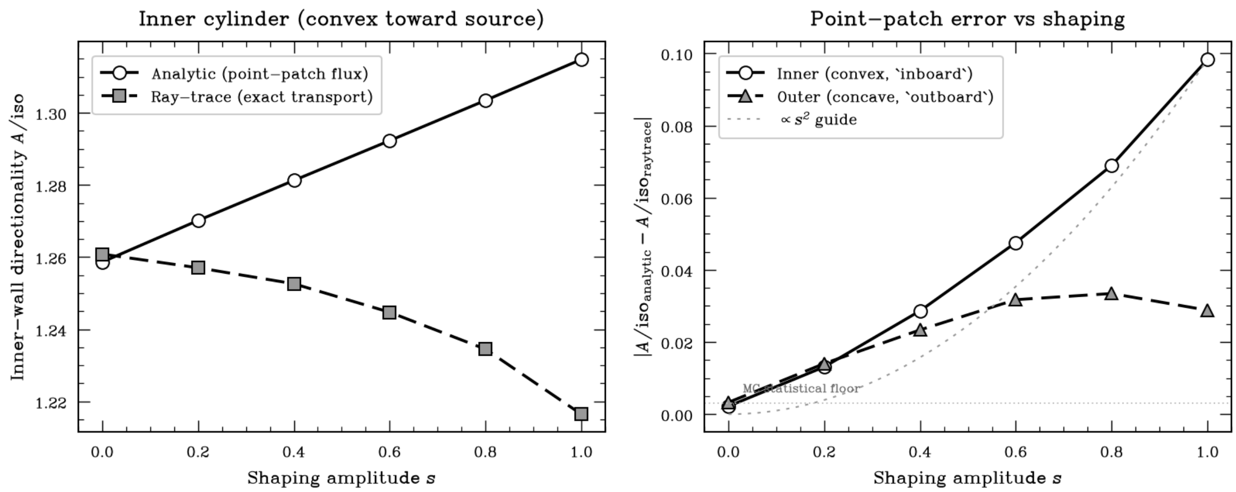

Figure 3: Single-filament demonstration that the inboard residual is the analytic point-patch limit. Left: inner-wall directionality A/iso from the analytic point-patch flux integral and from exact ray-tracing of the same births, versus the shaping amplitude s; coincident for an axisymmetric source ($s = 0$) and diverging as s grows. Right: the analytic↔ray-trace discrepancy versus s for the convex inner (inboard) and concave outer (outboard) cylinders, with an s^2 guide.

The one geometric approximation in the analytic NWL is that each wall patch is a point: the emission kernel is evaluated along the single line of sight from the source to the patch centre, whereas a patch on a curved wall subtends a finite solid angle over which the kernel varies, and the ray-traced Monte-Carlo resolves it. To exhibit this in isolation we strip the problem to a single shaped filament with a tunable shaping amplitude s, emitting the kernel about a purely toroidal field onto bare inner and outer cylinders, and evaluate the wall directionality two ways: the analytic point-patch flux integral, and an exact ray/surface intersection of the same sampled births. Here the sampler, field, and occlusion are all trivially correct, so s is the only variable. The result (Fig. 3) is decisive: at $s = 0$ the point-patch integral is exact (agreement to 0.002, the statistical floor), and as s increases the discrepancy grows as s^2, about three times faster on the convex inner cylinder than on the concave outer, reaching $\sim 10\%$ on the inner wall at the strongest shaping. This reproduces both the inboard-specific location and the shaping scaling of the full-equilibrium residual, and identifies it unambiguously as the point-patch treatment of the flux integral on a strongly curved wall—a property of the analytic method, with the geometry-exact, sampler-exact Monte-Carlo source as the reference. It is precisely the strongly shaped, strongly curved regime that motivates a transport calculation, and to which we now turn.

7 Transport on a realistic wall: scattering dilutes the steering

The free-streaming cross-checks certify the source; the reason to have a Monte-Carlo source is what the analytic method cannot do, and the first of these is neutron transport. To obtain a first, clean estimate of how much scattering erodes the geometric steering, we isolate the transport effect by measuring it in the same axisymmetric square-torus geometry used for validation—holding the wall fixed and changing only the material, so that any departure from the free-streaming steering is attributable to scattering alone rather than to a change of shape, and the radially layered blanket gives every neutron a well-defined path through the layers for an unambiguous energy and breeding

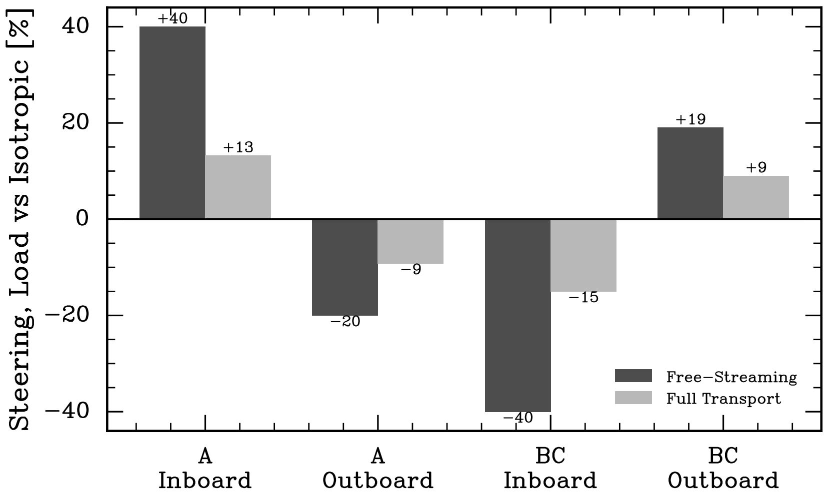

Figure 4: Scattering dilutes the geometric steering by $\sim 3\times$. The free-streaming inboard/outboard contrast (dark) versus the same quantity transported through a real layered blanket with scattering on (light), for the perpendicular (A) and parallel (B/C) modes. The $\pm 40\%$ geometric contrast realizes as only ± 13–15% at the wall.

accounting. Replacing the void interior and vacuum walls with an ARC-class layered blanket—a 1 mm tungsten first wall, 1 cm of Fe-9Cr structural steel, a 1 cm beryllium multiplier, and 40 cm of FLiBe breeder—turns the first-flight NWL into the transported flux and heating that a reactor blanket actually sees. This box-torus estimate isolates the dilution factor; the definitive transported load on the conformal geometry, whose free-streaming limit is validated above (Fig. 6), follows by running the identical source and blanket on the DAGMC conformal wall, and is expected to carry the same dilution. The polarized angular emission enters this calculation only through the source; everything downstream is standard neutron transport, so the anisotropy is carried into the volumetric loads without further approximation. As an anchor, reducing every layer density by 10^{-4} (mean free path $\gg$ device) recovers the validated free-streaming steering to within statistics—the A mode raising the inboard load by $+40.0\%$ and lowering the outboard by -19.5%, against the free-streaming $+39\%/-21\%$—so the layered model and the compiled source are correct with the full geometry.

The transported result is that *scattering dilutes the steering by a factor of* ~ 3 (Fig. 4). With the real FLiBe/Be blanket, neutrons that enter the wall scatter back into the cavity as a diffuse, direction-blind flux, washing out the directional contrast: the A mode's inboard enhancement falls from $+40\%$ to $+13.2\%$ and its outboard suppression from -20% to -9.2%, while the B/C mode's inboard relief falls from -40% to -15.0% and its outboard rise from $+19\%$ to $+9.0\%$. The parallel mode still relieves the inboard center stack, but by $\sim 15\%$, not the $\sim 41\%$ the geometric picture suggests. This is exactly the correction the free-streaming and analytic methods structurally cannot provide, and it is a caution for the field: geometric and analytic steering numbers, ours included, are optimistic by a factor of ~ 3 once a scattering blanket is present. (The energy bookkeeping confirms an active blanket: the FLiBe absorbs 91.4% of the deposited energy and the tungsten first wall only $\sim 0.1\%$, with total heating $0.87\times$ the birth energy and a down-scattered keV tail that is absent in the free-streaming model and that drives the low-energy breeding.)

Table 3: Absolute quantities at fixed fuel density, relative to unpolarized fuel = 1 (scattering on). The A mode's rate boost compounds with its inboard steering; the B mode buys steering at no rate cost; the TBR per neutron is mode-independent.

mode	rate η	fusion rate	center-stack load	outboard load	TBR/neutron
unpolarized	0.667	1.00	1.00	1.00	1.149
A	1.000	1.50	**1.70**	1.36	1.144
B	0.667	1.00	**0.85**	1.09	1.155
C	0.333	0.50	0.42	0.55	1.155

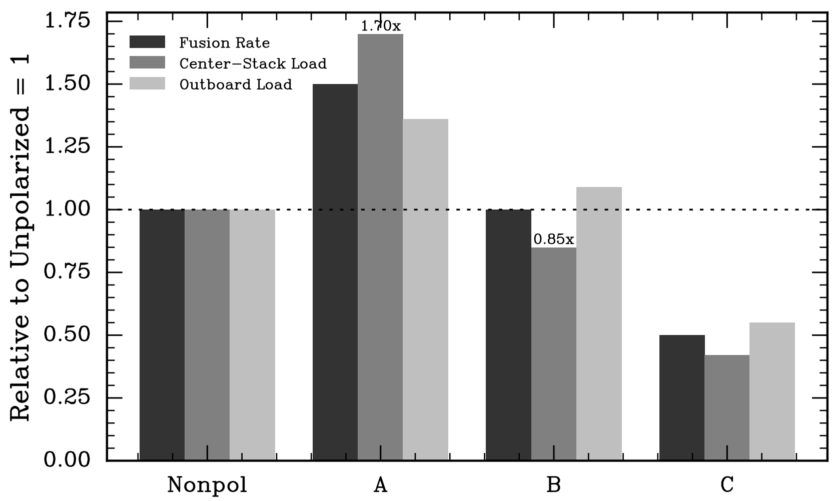

Figure 5: The rate-versus-steering trade-off (relative to unpolarized = 1). The A mode buys power but compounds the center-stack load to 1.70×; the B mode buys inboard relief (0.85×) at no rate cost; the C mode halves everything. The rate boost lives in a different spin state than the center-stack steering.

8 The rate-versus-steering trade-off

The cross-checks above held the fusion rate fixed to isolate directionality. Restoring it at fixed fuel density—the neutron yield scales with the rate factor $\eta = a + \frac{2}{3}b + \frac{1}{3}c$ of Eq. (2), which carries the Kulsrud reactivity enhancement [4] the constant-rate comparisons normalized away—separates what the three polarization modes buy (Fig. 5, Table 3). The A mode delivers +50% fusion power and +49% absolute tritium, but because its emission is perpendicular to $\hat{\boldsymbol{B}}$ it steers neutrons toward the inboard midplane, so the center-stack load *compounds* to 1.70× (rate times inboard steering)—excellent for power, hard on the center stack. The B mode carries the same power and breeding as unpolarized fuel but relieves the center-stack load to 0.85×: it buys the directional benefit at no rate penalty, and is the sweet spot for wall-load control. The C mode halves both power and breeding but gives the lowest absolute center-stack load (0.42×), worthwhile only if the center-stack load is the single binding constraint. Crucially, the tritium breeding ratio *per neutron* is nearly mode-independent (1.144–1.155), so the rate boost breeds proportionally more tritium and self-sufficiency is preserved: polarization redistributes and rescales the load without endangering breeding adequacy.

9 Conformal-wall loads: a strong local, weak global actuator

The steering results above use a box-torus for a clean scattering and rate accounting; we now place the polarized source on the conformal first wall of a real stellarator, the geometry a reactor would actually build. For each of fifteen quasi-symmetric QUASR equilibria we take the VMEC last-closed surface, offset it outward by $0.30\,a$ along its poloidal normal, mesh it as a DAGMC first wall [12], and free-stream the polarized `StellaratorSource` onto it, capturing wall crossings and binning them into a (ϑ, ϕ) neutron-load map (three million crossings per mode). Because the load is exactly linear in the polarization parameter a_2, the perpendicular and parallel maps satisfy $\mathrm{NWL}_\perp + \mathrm{NWL}_\| = 2\,\mathrm{NWL}_{\mathrm{unpol}}$ bin by bin—an exact self-consistency identity requiring no analytic reference, which the transported maps confirm to the few-percent Monte-Carlo level and which we use as an in-situ correctness check on the whole equilibrium-to-map pipeline.

Before drawing physics from these maps we validate the source free-streaming *on the conformal geometry itself*, so that the cross-check is not limited to the axisymmetric square torus of §5. On the gentle QA (device 59509, $\varepsilon_{\mathrm{eff}} \approx 0.43$) we evaluate the analytic point-patch free-streaming directionality A/iso and B/iso on the same (ϑ, ϕ) conformal wall grid the ray-traced OpenMC source was binned onto, using the local field $\hat{\boldsymbol{B}}$ at each source voxel and front-facing visibility. The two agree closely (Fig. 6): the per-bin correlation is 0.91, the global mean directionality matches to $\leq 1.2\%$ (A/iso analytic 1.027 vs Monte-Carlo 1.018; B/iso 0.973 vs 0.985), and by region the agreement is 1.6%/2.3% inboard and 0.5%/0.6% outboard, with the expected sign—mode A enhances the inboard load and suppresses the outboard, mode B/C the mirror. The source therefore reproduces the free-streaming NWL on the real conformal wall, not only in the axisymmetric limit, and the residual is the same shaping-driven point-patch effect isolated in §6 (small here, at $\varepsilon_{\mathrm{eff}} \approx 0.43$; it grows on the strongly shaped devices, which is why they are Monte-Carlo-only).

Scanning $a_2 \in [-1, 1]$ for the polarization that minimizes the peaking factor $\mathrm{PF} = \max \mathrm{NWL}/\overline{\mathrm{NWL}}$, we find the optimum is device-specific (parallel for most devices, perpendicular for a few) but the achievable global flattening is uniformly modest across the fifteen: peaking-optimal polarization lowers the peaking factor by 0–16%, with a median of only 4% (for the most improved device, $1.78 \to 1.49$; Fig. 7), even though the same polarization commands 40–70% swings in the *local* load between the perpendicular and parallel extremes. The reason is structural: the peaking factor is set by a narrow near-field spike—the wall's closest approach to the plasma, a few tenths to a few percent of the wall area—and angular redistribution reshapes the broad load but cannot move that spike. Polarization is therefore a strong local actuator and a weak global flattener of the wall load. The proper use of SPF steering is not to reduce global peaking but to tailor the fluence locally, for instance to relieve a lightly shielded port or a maintenance-critical panel; the global peaking is a geometry problem that wall shaping, not polarization, must solve.

It is natural to ask whether the modest, device-specific benefit can be predicted from a cheap geometric descriptor, so that the transport calculation could be skipped in a design scan. Across the fifteen devices it cannot (Fig. 8). Pure-plasma shape metrics—the non-axisymmetric shaping amplitude of the boundary and a nematic order parameter of its normals—do not correlate with the steering benefit (Spearman $\rho = -0.19$ and $+0.40$ respectively, neither significant at $n = 15$), consistent with the line-of-sight integration acting as a low-pass filter that removes the high-harmonic shape content from the wall signal. The best predictor is a transport-free isotropic $1/D^2$ peaking proxy—essentially the available headroom, since a device with a larger geometric hot spot has more to gain—but even it is only borderline ($\rho = +0.53$, $p \approx 0.05$), and a spuriously strong correlation seen on the first five devices ($\rho = +0.90$) regressed toward this value as the sample grew. The practical conclusion reinforces the paper's thesis: the polarization steering benefit is weak, device

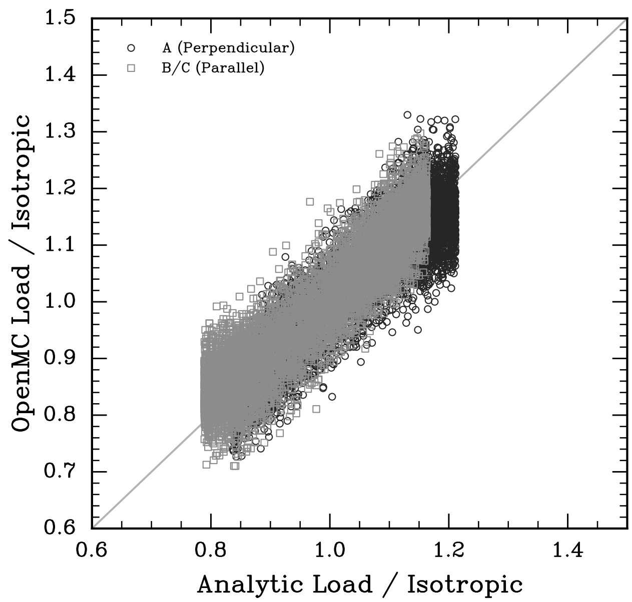

Figure 6: Free-streaming cross-check *on the conformal wall* (device 59509): analytic point-patch NWL directionality versus the ray-traced OpenMC `StellaratorSource`, per wall bin, for modes A (perp) and B/C (par). The two agree to 0.5–2.3% by region (per-bin correlation 0.91), validating the source free-streaming on the real three-dimensional geometry, not only on the axisymmetric square torus.

specific, and not forecastable from plasma shape, so it must be evaluated per device by the transport calculation the source provides.

10 The analytic companion and the division of labour

The transport results are cross-checked by, and delimit the validity of, a differentiable analytic framework for polarized NWL developed in parallel. Its reusable contributions are three. First, an analytic *singularity subtraction* for the near-singular free-streaming integral: because the squared source-to-wall distance is a trigonometric polynomial, the near-singularity's location, depth, and curvature are available in closed form, so one can subtract an exactly integrable model of the grazing peak and integrate the smooth remainder with twelve quadrature nodes where fixed Gauss–Legendre needs more than six thousand, all while remaining automatically differentiable—a construction that sharpens the singularity-swap approach of af Klinteberg and Barnett [9] by obtaining the peak parameters in closed form rather than by root-finding. Second, the only *polarized* NWL kernels in the analytic literature, riding a differentiable hybrid quadrature that yields exact geometry gradients—a normalization correction to the second-order polarized kernel, restoring the unit-field constraint that the leading derivation drops, advances the polarized series from second to third order. Third, a characterization of the effective shaping parameter $\varepsilon_{\mathrm{eff}}$ with two distinct thresholds—a series-convergence radius near $\frac{1}{2}$ and a larger grazing/quadrature breakdown near 2—which quantifies why good quasi-symmetry and geometric near-axisymmetry anti-correlate, and thereby why the strongly shaped published QAs of §5 sit in the Monte-Carlo-only regime.

The companion independently reproduces the central result of §9: a single global polarization flattens the full-wall load by under a factor of two, the same weak-global-lever conclusion the

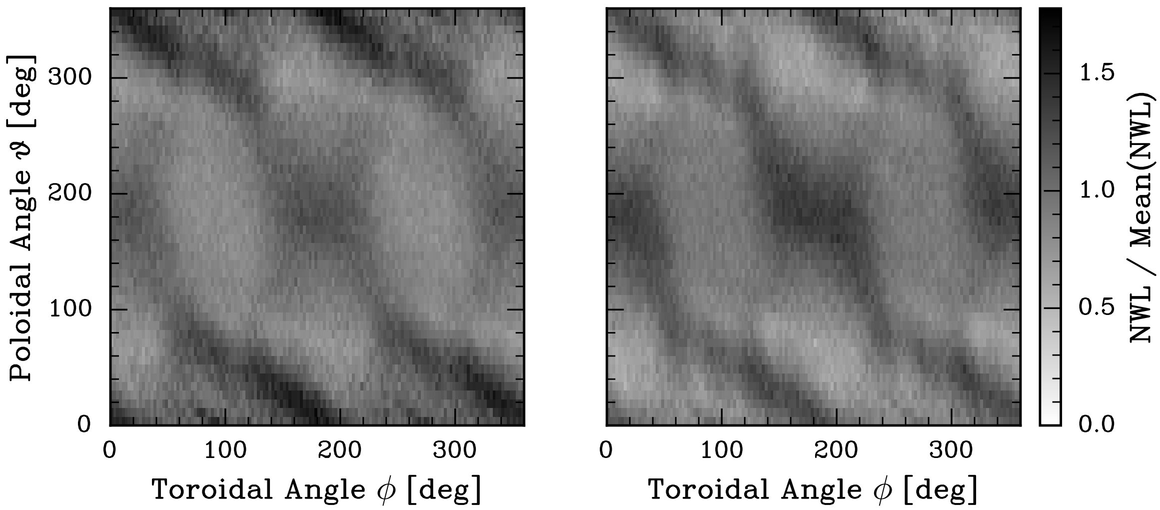

Figure 7: Neutron wall load NWL/$\overline{\mathrm{NWL}}$ on the DAGMC conformal first wall of QUASR device 886079 (real free-streaming `StellaratorSource` transport): unpolarized (left) versus peaking-optimal polarization $a_2 = +0.82$ (right). The peaking factor falls only from 1.78 to 1.49: the near-field hot spot is not steerable, though the broad load is visibly redistributed.

transport reaches, from an entirely separate calculation. Its native regime is small periodic non-axisymmetry, which maps most honestly not onto strongly shaped stellarators but onto tokamak toroidal-field ripple, where the wall signature of the ripple is exponentially suppressed as e^{-Na/R_0} because line-of-sight integration acts as a low-pass filter on high-harmonic source structure. The two methods are therefore not competitors but a division of labour: the closed, differentiable analytic form where the boundary is weakly shaped and gradients drive optimization, and the present source where the shaping is strong, where the wall is conformal, or where the neutron transport that the free-streaming methods cannot perform is what a reactor design actually needs.

11 Limitations

Several bounds on the present results should be stated. The scattering and rate-trade-off studies (§7–8) use a layered box-torus blanket rather than the conformal stellarator wall of §9; the two are used at the fidelity that isolates their respective effects—the box-torus for a clean volumetric energy and breeding accounting, the conformal wall for the geometric load pattern—but a unified conformal, scattering run is the natural next step and is expected to inherit the $\sim 3\times$ dilution. The steering results are monoenergetic at 14.1 MeV; a Ballabio-broadened source is available and does not affect the angular physics. The analytic companion is first-flight and convex-patch, its grazing-robust path is validated in the unpolarized channel and holds for the polarized channel by construction rather than by test, and its reverse-mode gradients through the special-function engine do not yet just-in-time compile. Finally, the conformal-wall study covers fifteen equilibria—enough to establish that the steering benefit is weak and not forecastable from plasma shape, but the highly shaped quasi-helical configurations that fail to converge in the fixed-boundary VMEC solve are under-represented, so the device sample is biased toward the more tractable quasi-axisymmetric class.

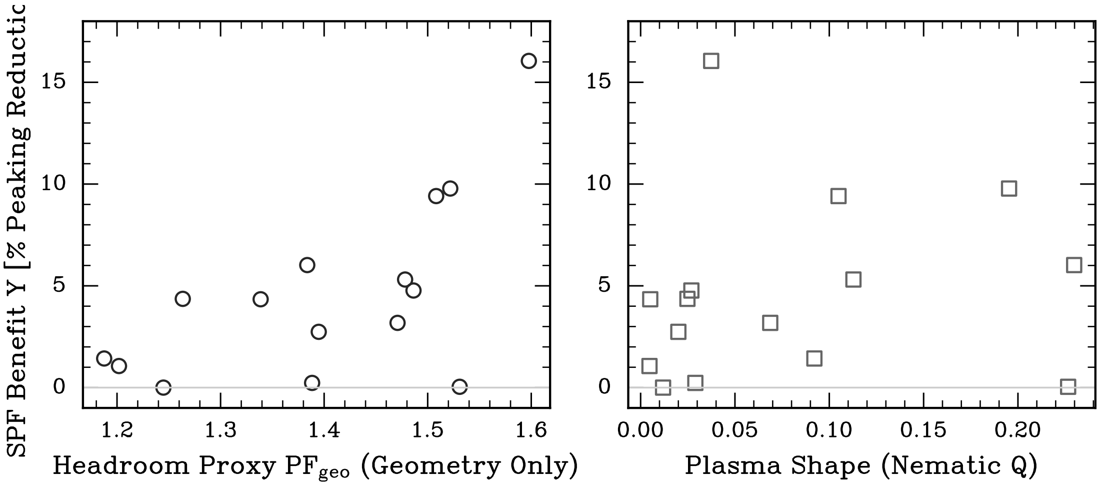

Figure 8: The polarization steering benefit is weak and not forecastable from plasma shape ($n = 15$ conformal-wall devices). Left: the benefit correlates only weakly with a transport-free isotropic peaking (headroom) proxy ($\rho = +0.53$, borderline at $n = 15$). Right: it does not correlate with a pure-plasma shape metric ($\rho = +0.40$, not significant). The steering benefit must be evaluated per device by transport.

Data availability. All numbers, tables, and figures in this paper are reproducible from a self-contained archive: the frozen transport outputs (the conformal-wall (ϑ, ϕ) load maps, the scattering and rate ledgers, the fifteen-device predictor table) are stored with a provenance manifest naming the run that produced each, and one figure script per figure reads only that archive. The native source, the VMEC producer with its rigor gates, and the analysis and figure scripts are released with the archive.

12 Summary

We have built and verified the first native, on-the-fly spin-polarized fusion neutron source in a production Monte-Carlo transport code, realized as an equilibrium-rigorous `StellaratorSource`—the first stellarator source, and the first polarized stellarator source, in OpenMC. The emission kernel separates cleanly into a birth-direction distribution, sampled exactly by stratification with inverse-CDF draws cross-checked against rejection, and a scalar rate factor; the sampler reproduces the normalized polarized kernels to 0.3% with unit mean, and the transported free-streaming loads reproduce the analytic NWL to $\leq 1.8\%$ on a real published QA equilibrium, the one 4–6% inboard residual identified by construction as the analytic point-patch limit rather than a defect of the source. Turning on transport, we find that scattering dilutes the geometric steering by a factor of ~ 3—a correction only a transport calculation can supply; restoring the reactivity, we find that the +50%-rate mode compounds the center-stack load to 1.70× while the equal-rate parallel mode relieves it to 0.85× at no cost to power or to the tritium breeding ratio; and on conformal first walls of real stellarators we find that peaking-optimal polarization lowers the peaking factor only modestly because the load peak is an unsteerable near-field spike, so polarization is a strong local but weak global actuator—a conclusion an independent differentiable analytic method reaches as well. The source is available for both tokamak- and stellarator-class equilibria and provides a verified, honest foundation for polarized-fuel reactor neutronics.

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